

WESTCHESTER COUNTY AP, NY, USA

WMO: 725037

Lat: 41.062N

Lon: 73.705W

Elev: 112

StdP: 99.99

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 94745

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.9	-10.6	-22.2	0.5	-11.4	-19.6	0.7	-8.8	12.7	-6.0	11.6	-5.9	5.3	310	0.479

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	31.9	23.0	30.2	22.2	28.8	21.5	24.6	29.2	23.8	27.9	23.0	26.6	3.6	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	18.0	26.4	22.5	17.5	25.8	22.0	16.9	25.3	74.9	29.2	71.6	27.9	68.7	26.6	27.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.7	8.9	7.9	DB	-15.8	34.7	3.2	1.9	-18.1	36.1	-20.0	37.2	-21.8	38.2	-24.1	39.6	(4)
(5)			WB	-16.7	26.1	2.9	0.9	-18.9	26.7	-20.6	27.2	-22.2	27.7	-24.4	28.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.3	-1.3	0.0	3.9	9.9	15.3	20.4	23.5	22.6	18.8	12.7	6.9	2.0
	DBStd	9.55	5.64	5.12	4.84	4.33	3.96	3.47	2.79	2.72	3.49	4.16	4.48	4.78
	HDD10.0	1266	350	282	196	52	3	0	0	0	0	21	113	251
	HDD18.3	3021	607	514	446	254	112	19	1	2	36	182	343	505
	CDD10.0	1741	1	1	8	50	168	313	417	391	264	103	21	4
	CDD18.3	453	0	0	0	2	19	82	159	134	51	6	0	0
	CDH23.3	3263	0	0	2	34	178	597	1253	903	270	26	0	0
	CDH26.7	937	0	0	0	8	45	166	409	252	55	3	0	0
Wind	WSAvg	3.4	4.0	4.0	4.2	3.8	3.2	2.9	2.6	2.5	2.8	3.1	3.6	3.8
Precipitation	PrecAvg	1270	88	73	114	109	97	116	108	118	123	116	101	112
	PrecMax	1558	183	186	197	288	198	210	203	239	346	406	171	171
	PrecMin	1007	44	26	21	44	17	34	33	38	35	17	32	31
	PrecStd	161	35	34	46	52	41	51	46	46	80	86	37	36
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.9	16.1	21.9	27.7	30.9	32.6	34.7	33.0	31.0	26.3	20.0	17.2
		MCWB	11.4	10.9	13.7	17.3	20.5	23.3	23.8	24.2	22.4	19.0	15.3	13.1
	2%	DB	12.3	12.2	17.2	23.0	27.6	30.3	32.3	31.0	28.0	23.2	17.6	13.8
		MCWB	10.3	8.5	11.0	14.5	19.2	22.1	23.3	22.9	20.9	18.4	14.0	11.4
	5%	DB	9.1	9.6	13.8	20.0	25.1	28.6	30.5	29.2	26.3	21.2	15.9	11.5
		MCWB	6.8	6.3	8.9	12.7	17.5	21.3	22.5	21.9	20.0	17.1	12.9	9.5
	10%	DB	6.6	7.3	11.2	17.3	22.5	26.9	28.9	27.8	24.6	19.3	14.0	9.1
		MCWB	4.3	4.1	7.1	11.0	16.2	20.1	21.8	21.3	19.5	15.8	11.1	6.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.1	12.9	15.2	18.4	22.2	24.6	25.9	25.7	24.1	21.6	17.3	14.6
		MCDB	13.9	14.6	20.1	25.1	28.3	30.2	32.0	30.4	27.9	23.6	18.5	16.0
	2%	WB	10.6	9.3	12.3	16.1	20.6	23.5	24.8	24.5	22.9	19.7	15.4	12.0
		MCDB	11.8	11.1	15.2	20.9	25.8	28.4	29.7	28.5	25.6	22.3	16.8	13.3
	5%	WB	7.0	6.5	9.7	14.3	19.2	22.4	23.9	23.6	22.0	18.1	13.6	10.0
		MCDB	8.7	8.7	13.1	18.1	23.3	26.9	28.3	27.1	24.6	20.3	15.3	11.5
	10%	WB	4.5	4.3	7.7	12.3	17.6	21.4	23.0	22.8	20.9	16.5	11.7	7.1
		MCDB	6.5	7.1	10.4	15.7	21.1	25.1	26.7	26.0	23.3	18.8	13.6	8.6
Mean Daily Temperature Range	5% DB	MDBR	7.5	8.3	8.7	9.9	9.9	9.5	9.2	9.0	9.1	8.8	8.0	7.0
		MCDBR	10.0	11.4	12.5	14.1	13.5	11.6	10.8	10.4	10.6	10.9	10.1	9.7
		MCWBR	9.0	8.7	8.2	7.6	6.7	5.4	4.4	4.4	5.2	6.5	7.8	8.7
	5% WB	MCDBR	9.5	10.6	11.4	12.2	11.7	10.1	9.1	8.7	8.5	9.0	8.8	9.2
		MCWBR	9.3	9.2	8.8	7.8	6.9	5.5	4.5	4.4	5.3	6.8	8.1	9.1
Clear-Sky Solar Irradiance	taub	0.309	0.321	0.350	0.374	0.420	0.475	0.485	0.465	0.392	0.354	0.325	0.309	
	taud	2.447	2.423	2.408	2.377	2.273	2.159	2.143	2.188	2.366	2.487	2.524	2.517	
	Ebn at Noon	846	890	900	897	859	810	797	801	843	843	825	820	
	Edh at Noon	78	93	105	117	133	149	150	140	109	87	73	68	
All-Sky Solar Radiation	RadAvg	1.84	2.73	3.73	4.77	5.32	5.82	5.96	5.26	4.32	2.96	2.07	1.55	
	RadStd	0.15	0.26	0.30	0.49	0.53	0.54	0.31	0.34	0.39	0.30	0.18	0.15	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+101	+57	(46)
(47) Regional (63 neighbors)		+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+114	+64	(47)

Nomenclature: See separate page